

The screenshot shows a VS Code editor with a project named 'inteligencia-artificial'. The Explorer panel on the left shows a directory structure with files like '01_romania_map.py', '02_breadth_first_search.py', and '03_uniform_cost_search.py'. The main editor displays the code for '02_breadth_first_search.py'. The code imports necessary modules, defines a search problem, and implements a breadth-first search algorithm. The output window at the bottom shows the results of running the program: 'Algorithm: Breadth-first search', 'Problem: Sibiu -> Eforie', 'Status: success', 'Path: Sibiu -> Fagaras -> Bucharest -> Urziceni -> Hirsova -> Eforie', 'Depth: 5 roads', 'Cost: 579 km', 'Expanded: 16 nodes', 'Generated: 40 nodes', and 'Frontier: max size 5'.

```
02_breadth_first_search.py X
Búsqueda no informada > project > 02_breadth_first_search.py > main
1 #!/usr/bin/env python3
2 """Program 2a: breadth-first search on the Romania map."""
3
4 from __future__ import annotations
5
6 import sys
7 from pathlib import Path
8
9 ROOT = Path(__file__).resolve().parent
10 sys.path.insert(0, str(ROOT))
11
12 from search.bfs import breadth_first_search # noqa: E402
13 from search.cli import make_problem, print_result, search_parser # noqa: E402
14
15
16 def main() -> None:
17     parser = search_parser("Breadth-first search: fewest roads, not fewest km.")
18     args = parser.parse_args()
19     problem = make_problem(args.start, args.goal)
20     result = breadth_first_search(problem)
21     print_result("Breadth-first search", problem, result)
22
23
24 if __name__ == "__main__":
25     main()
26
Problems Output Debug Console Terminal Ports
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project> python 02_breadth_first_search.py --from-city Sibiu --to Eforie
Algorithm: Breadth-first search
Problem: Sibiu -> Eforie
Status: success
Path: Sibiu -> Fagaras -> Bucharest -> Urziceni -> Hirsova -> Eforie
Depth: 5 roads
Cost: 579 km
Expanded: 16 nodes
Generated: 40 nodes
Frontier: max size 5
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project>
```

The screenshot shows a VS Code editor with the same project. The Explorer panel shows the directory structure, and the main editor displays the code for '03_uniform_cost_search.py'. The code imports necessary modules, defines a search problem, and implements a uniform-cost search algorithm. The output window at the bottom shows the results of running the program: 'Algorithm: Uniform-cost search', 'Problem: Sibiu -> Eforie', 'Status: success', 'Path: Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest -> Urziceni -> Hirsova -> Eforie', 'Depth: 6 roads', 'Cost: 547 km', 'Expanded: 17 nodes', 'Generated: 43 nodes', and 'Frontier: max size 6'.

```
03_uniform_cost_search.py X
Búsqueda no informada > project > 03_uniform_cost_search.py > ...
1 #!/usr/bin/env python3
2 """Program 2b: uniform-cost search on the Romania map."""
3
4 from __future__ import annotations
5
6 import sys
7 from pathlib import Path
8
9 ROOT = Path(__file__).resolve().parent
10 sys.path.insert(0, str(ROOT))
11
12 from search.cli import make_problem, print_result, search_parser # noqa: E402
13 from search.ucs import uniform_cost_search # noqa: E402
14
15
16 def main() -> None:
17     parser = search_parser("Uniform-cost search: cheapest path in km.")
18     args = parser.parse_args()
19     problem = make_problem(args.start, args.goal)
20     result = uniform_cost_search(problem)
21     print_result("Uniform-cost search", problem, result)
22
23
24 if __name__ == "__main__":
25     main()
26
Problems Output Debug Console Terminal Ports
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project> python 03_uniform_cost_search.py --from-city Sibiu --to Eforie
Algorithm: Uniform-cost search
Problem: Sibiu -> Eforie
Status: success
Path: Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest -> Urziceni -> Hirsova -> Eforie
Depth: 6 roads
Cost: 547 km
Expanded: 17 nodes
Generated: 43 nodes
Frontier: max size 6
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project>
```

```
03_uniform_cost_search.py X
Búsqueda no informada > project > 03_uniform_cost_search.py > ...
1 #!/usr/bin/env python3
2 """Program 2b: uniform-cost search on the Romania map."""
3
4 from __future__ import annotations
5
6 import sys
7 from pathlib import Path
8
9 ROOT = Path(__file__).resolve().parent
10 sys.path.insert(0, str(ROOT))
11
12 from search.cli import make_problem, print_result, search_parser # noqa: E402
13 from search.ucs import uniform_cost_search # noqa: E402
14
15
16 def main() -> None:
17     parser = search_parser("uniform-cost search: cheapest path in km.")
18     args = parser.parse_args()
19     problem = make_problem(args.start, args.goal)
20     result = uniform_cost_search(problem)
21     print_result("Uniform-cost search", problem, result)
22
23
24 if __name__ == "__main__":
25     main()
26
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project> python 04_depth_first_search.py --from-city Sibiu --to Eforie
Algorithm: Depth-first search
Problem: Sibiu + Eforie
Status: success
Path: Sibiu + Arad + Timisoara + Lugoj + Mehadia + Drobeta + Craiova + Pitesti + Bucharest + Urziceni + Hirsova + Eforie
Depth: 11 roads
Cost: 1142 km
Expanded: 12 nodes
Generated: 32 nodes
Frontier: max size 6
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project>
```

```
03_uniform_cost_search.py X
Búsqueda no informada > project > 03_uniform_cost_search.py > ...
1 #!/usr/bin/env python3
2 """Program 2b: uniform-cost search on the Romania map."""
3
4 from __future__ import annotations
5
6 import sys
7 from pathlib import Path
8
9 ROOT = Path(__file__).resolve().parent
10 sys.path.insert(0, str(ROOT))
11
12 from search.cli import make_problem, print_result, search_parser # noqa: E402
13 from search.ucs import uniform_cost_search # noqa: E402
14
15
16 def main() -> None:
17     parser = search_parser("uniform-cost search: cheapest path in km.")
18     args = parser.parse_args()
19     problem = make_problem(args.start, args.goal)
20     result = uniform_cost_search(problem)
21     print_result("Uniform-cost search", problem, result)
22
23
24 if __name__ == "__main__":
25     main()
26
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project> python 05_depth_limited_search.py --from-city Sibiu --to Eforie --limit 2
Generated: 32 nodes
Frontier: max size 6
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project> python 05_depth_limited_search.py --from-city Sibiu --to Eforie --limit 2
Algorithm: Depth-limited search
Problem: Sibiu + Eforie
Status: cutoff
Detail: limit=2
Expanded: 5 nodes
Generated: 15 nodes
Frontier: max size 5
PS C:\Users\andro\inteligencia-artificial\Búsqueda no informada\project>
```

```
File Edit Selection View Go Run Terminal Help
Inteligencia-artificial

03_uniform_cost_search.py X
Búsqueda no informada > project > 03_uniform_cost_search.py > ...
1 #!/usr/bin/env python3
2 """Program 2b: uniform-cost search on the Romania map."""
3
4 from __future__ import annotations
5
6 import sys
7 from pathlib import Path
8
9 ROOT = Path(__file__).resolve().parent
10 sys.path.insert(0, str(ROOT))
11
12 from search.cli import make_problem, print_result, search_parser # noqa: E402
13 from search.ucs import uniform_cost_search # noqa: E402
14
15
16 def main() -> None:
17     parser = search_parser("uniform-cost search: cheapest path in km.")
18     args = parser.parse_args()
19     problem = make_problem(args.start, args.goal)
20     result = uniform_cost_search(problem)
21     print_result("Uniform-cost search", problem, result)
22
23
24 if __name__ == "__main__":
25     main()
26
```

Problems Output Debug Console Terminal Ports

PS C:\Users\andro\Inteligencia-artificial\Búsqueda no informada\project> python 05_depth_limited_search.py --from-city Sibiu --to Eforie --limit 5

Algorithm: Depth-limited search
Problem: Sibiu -> Eforie
Status: success
Detail: limit=5
Path: Sibiu -> Fagaras -> Bucharest -> Urziceni -> Hirsova -> Eforie
Depth: 5 roads
Cost: 579 km
Expanded: 15 nodes
Generated: 34 nodes
Frontier: max size 8

```
File Edit Selection View Go Run Terminal Help
Inteligencia-artificial

03_uniform_cost_search.py X
Búsqueda no informada > project > 03_uniform_cost_search.py > ...
1 #!/usr/bin/env python3
2 """program 2b: uniform-cost search on the Romania map."""
3
4 from __future__ import annotations
5
6 import sys
7 from pathlib import Path
8
9 ROOT = Path(__file__).resolve().parent
10 sys.path.insert(0, str(ROOT))
11
12 from search.cli import make_problem, print_result, search_parser # noqa: E402
13 from search.ucs import uniform_cost_search # noqa: E402
14
15
16 def main() -> None:
17     parser = search_parser("uniform-cost search: cheapest path in km.")
18     args = parser.parse_args()
19     problem = make_problem(args.start, args.goal)
20     result = uniform_cost_search(problem)
21     print_result("Uniform-cost search", problem, result)
22
23
24 if __name__ == "__main__":
25     main()
26
```

Problems Output Debug Console Terminal Ports

PS C:\Users\andro\Inteligencia-artificial\Búsqueda no informada\project> python 06_iterative_deepening_search.py --from-city Sibiu --to Eforie

Algorithm: Iterative deepening search
Problem: Sibiu -> Eforie
Status: success
Detail: last limit=5
Path: Sibiu -> Fagaras -> Bucharest -> Urziceni -> Hirsova -> Eforie
Depth: 5 roads
Cost: 579 km
Expanded: 53 nodes
Generated: 143 nodes
Frontier: max size 8